SIM1002 Calculus I
Tutorial 9
Integrals

Exercise 5.2

Sigma Notation

Evaluate the sums in Exercises 1-3.

1. $\sum_{k=1}^{5} \frac{\pi k}{15}$

2. $\sum_{k=1}^{7} k(2k+1)$

3. (a) $\sum_{k=9}^{36} k$ (b) $\sum_{k=3}^{17} k^2$ (c) $\sum_{k=18}^{71} k(k-1)$

4. $\sum_{k=2}^{20} [\sin(k-1) - \sin k]$

5. $\sum_{k=1}^{40} \frac{1}{k(k+1)}$

Exercise 5.3

Using the Definite Integral Rules

1. Suppose that $\int_{-3}^{0} g(t)\, dt = \sqrt{2}$. Find

   (a) $\int_{0}^{-3} g(t)\, dt$ (b) $\int_{-3}^{0} g(u)\, du$

   (c) $\int_{-3}^{0} [-g(x)]\, dx$ (d) $\int_{-3}^{0} \frac{g(r)}{\sqrt{2}}\, dr$

2. Suppose that h is integrable and that $\int_{-1}^{1} h(r)\, dr = 0$ and $\int_{-1}^{3} h(r)\, dr = 6$. Find

   (a) $\int_{1}^{3} h(r)\, dr$ (b) $-\int_{3}^{1} h(u)\, du$

Using Known Areas to Find Integrals

In Exercises 3-5, graph the integrands and use known area formulas to evaluate the integrals.

3. $\int_{-4}^{0} \sqrt{16 - x^2}\, dx$

4. $\int_{-1}^{1} (1 - |x|)\, dx$

5. $\int_{-1}^{1} \left(1 + \sqrt{1 - x^2}\right) dx$

Use known area formulas to evaluate the integrals in Exercises 6-7.

6. $\int_a^b 3t\ dt,\ 0 < a < b$

7. $f(x) = 3x + \sqrt{1 - x^2}$, on (a) $[-1,0]$, (b) $[-1,1]$

Evaluating Definite Integrals

Evaluate the integrals in Exercises 8-12.

8. $\int_0^{\frac{\pi}{2}} \theta^2\, d\theta$

9. $\int_a^{\sqrt{3}} x\, dx$

10. $\int_0^{3b} x^2 dx$

11. $\int_0^{\sqrt{2}} \left(t - \sqrt{2}\right)\ dt$

12. $\int_1^0 (3x^2 + x - 5)\ dx$

Definite Integrals as Limits of Sums

Use the limit of a Riemann Sum to evaluate the definite integrals in Exercises 13-14.

13. $\int_{-1}^0 (x - x^2)\ dx$

14. $\int_0^1 (3x - x^3)\ \ dx$